

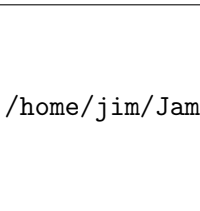
Section 4.4: Final Exam Review

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Friday 14th November, 2014



/home/jim/JamesAshe/MHU/images/MarsHillU

Schedule

<http://www.mhu.edu/academics/course-catalog/exam-schedule>

Final Exam	Date	Time
Section 07	Friday 5 th December	8-10am
Section 10	Wednesday 3 rd December	2:30-4:30pm

Chapter 1: Logic

- Words into symbols

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Example 4.4.1.

We do our homework or we do not miss class.

Chapter 1: Logic

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We do our homework or we do not miss class.

- Symbols into words

Chapter 1: Logic

- Words into symbols

Example 4.4.1.

We do our homework or we do not miss class.

- Symbols into words

Example 4.4.2.

Let h represent "We do our homework", and m represent we miss class". Symbolize: *If we miss class then we did not do our homework.*

Chapter 1: Logic

- Negation

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Example 4.4.3.

Negate: Some people are not going to have fun

Chapter 1: Logic

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- Construct a truth table

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Example 4.4.3.

Negate: Some people are not going to have fun

- Construct a truth table

Example 4.4.4.

Construct a truth table for $(p \vee q) \rightarrow \sim q$

Chapter 1: Logic

- Use a truth table to determine if an argument is valid.

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- Use a truth table to determine if an argument is valid.

Example 4.4.5.

m : You like money. 1. If you like food then you like money.
 f : You like food. 2. You do not like food.

Therefore, you do not like money.

		1	2	$1 \wedge 2$	$\neg$	Argument
f	m					
T	T					
T	F					
F	T					
F	F					

Chapter 2: Sets

- Basic set operations

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Example 4.4.6.

Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$, $C = \{e, f, g\}$, and $U = \{a, b, c, d, e, f, g\}$.

Chapter 2: Sets

- Basic set operations

Example 4.4.6.

Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$, $C = \{e, f, g\}$, and $U = \{a, b, c, d, e, f, g\}$.

- Find $A \cup B$

Chapter 2: Sets

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- Find $A \cup B$
- Find $A \cap B$
- Find $A \cap C$
- Find A'

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- Find A'
- Find $(A \cup B)'$

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- Find $A' \cap B'$
- Find $n(A)$
- Find $n(A \cup B)$

Chapter 2: Sets

- Venn Diagrams

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- Venn Diagrams

Example 4.4.7.

A class has 50 students. 20 are biology majors, 30 are math majors, and 10 are math and biology majors. Construct and Venn diagram and label the number of students in each region

Chapter 2: Counting

- Basic operations:

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Example 4.4.8.

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- $5!$

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Example 4.4.9.

A multiple choice test has 3 options for problems 1-3 and 4 options for problem 4-5. How many different ways can you select answers?

Chapter 2: Counting

Fundamental Principle of Counting:

- When things can be repeated.

- No repeats
- Order matters

- ${}_nP_k = \frac{n!}{(n-k)!}$

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How many ways can we choose 1st, 2nd, and 3rd place in a beauty contest with 5 contestants?

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Example 4.4.12.

There are 10 balls. 6 blue and 4 red. Four balls are randomly selected. Find the probability that at least 2 are blue.

Chapter 3: Probability

- $P(A) = \frac{n(A)}{n(U)}$

Find Simple Probability

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Example 4.4.13.

What is the probability that a $Q\spadesuit$ is drawn from a 52 card deck

Find Simple Probability

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- List of everything that can happen

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Example 4.4.14.

Two children are born; each is a girl or boy. Find the sample space

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Example 4.4.15.

What is the probability of having a boy and a girl?

Chapter 3: Probability

With Venn Diagrams

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Example 4.4.16.

A candy bag has 25% caramels, 40% chocolates, and 15% chocolate caramels. If you randomly grab a candy, find the probability of the following:

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- b. It is neither a caramel or chocolate candy.
- c. It is just a chocolate candy.

Chapter 3: Probability

Conditional Probability

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Conditional Probability

From a chart

Age	Obama	McCain	Other/NA
18-29	11.9%	5.8%	0.4%
30-44	15.1%	13.3%	0.6%
45-64	18.5%	18.1%	0.4%
65+	7.2%	8.5%	0.3%

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Example 4.4.18.

A car lot has 60% blue cars and the rest are red. Of the blue cars 25% are used and the rest are new. Of the red cars 80% are used and the rest are new.

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- Draw a tree diagram.
- Find the probability that a red car is new.
- Find the probability that a randomly selected car is red AND it is new.

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Descriptive

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